

Aryan Kumar Rai

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science (Specialization in AI & ML); CGPA: 8.18

Bhopal, Madhya Pradesh

Aug. 2024 – May 2028

DAV Public School

Class 12 – 85.8% | Class 10 – 90.8%

Jharsuguda, Odisha

2022 – 2024

EXPERIENCE

JSW JFE Steel Ltd. (Formerly JSW Sambalpur Steel Ltd.)

IT Department Intern

Sambalpur, Odisha

May 2026 – Jul 2026

- Owned an enterprise reporting system end-to-end – authentication, backend API, and database layer – for a portal serving 7 departments, as part of a 3-person team; took the project from ambiguous, paper-based requirements to a working production system with no prior spec.
- Designed a reliable 4-factor authentication scheme (JWT + bcrypt) with rate-limited endpoints (slowapi) and a full audit-log system, prioritizing correctness and traceability over shortcuts.
- Modeled department-isolated relational schemas from unstructured daily report sheets, debugging real production issues (SQLAlchemy identity-map collisions, IDENTITY column mismatches) under live deployment constraints.

PROJECTS

Lead Automation System | Python, FastAPI, Gemini AI, n8n, Celery, Redis, PostgreSQL | [GitHub](#) | [Demo](#)

- Built an async, fault-tolerant lead-processing pipeline (FastAPI + Celery + Redis) that classifies and enriches each lead end-to-end in under 3 seconds, with an async endpoint returning a task ID in under 50ms and 3-attempt auto-retry on failure.
- Designed idempotency checks that eliminated all duplicate database entries on webhook retries, and a keyword-based fallback classifier so the system degrades gracefully instead of failing when the AI API is unavailable.
- Implemented centralized logging and a global exception handler across all services for full traceability – treating reliability and observability as first-class requirements, not afterthoughts.

Department Wise Daily Report Portal | FastAPI, SQLAlchemy, MS SQL Server, JWT, bcrypt

- Reverse-engineered messy, department-specific data sheets into normalized, isolated relational schemas across 7 departments with zero cross-department data leakage.
- Built rate-limited, audit-logged authentication and role-based access control (Operator/Admin/Plant Admin/Developer) from scratch.

PharmaKrypt | React, Vite, Firebase, Tailwind CSS | [GitHub](#) | [Paper](#) | [Link](#)

- Built a medicine authentication system using cryptographic QR serialization and geofenced verification, validating each pack in under 2 seconds; enforced a one-scan consumption policy via Firestore atomic transactions, eliminating duplicate-scan counterfeits across all simulated SKUs.
- Co-authored a write-up of the system architecture, published in IJIRT (2026).

TECHNICAL SKILLS

- Languages:** C++ (DSA), Python, Java, SQL
- Backend & Systems:** FastAPI, Flask, SQLAlchemy, RESTful APIs, JWT, Celery, Redis, Asynchronous Systems
- Databases:** MS SQL Server, PostgreSQL, Firebase (Firestore)
- Cloud & Tools:** Docker, Google Cloud Platform, Git, GitHub Actions (basics), Postman
- AI-Assisted Development:** Daily use of Claude and AI coding tools for rapid prototyping, debugging, and code review to ship faster
- Concepts:** Data Structures & Algorithms, System Design (Fundamentals), OOP, Reliability & Observability

ACHIEVEMENTS & LEADERSHIP

- Competitive Programming:** Solved 134 problems on LeetCode (137 active days, 41-day max streak); following Striver's A2Z DSA Sheet in C++.
- 1st Runner-Up**, Summer of Codefest '25 Hackathon (VIT Bhopal) | [Certificate](#); participated in 10+ hackathons across 2024–2026, finalist in majority.
- Tech Team Lead, iCreate VITB:** Maintained technical infrastructure and automated event registration workflows used by 300+ participants.
- Publications:** “PharmaKrypt: Agentic AI-Powered Framework for Secure and Intelligent Medicine Verification” (IJIRT, 2026) | [Paper](#); “RubricAI: AI-Driven Automated Assignment Evaluation” (IJFMR, 2025) | [Paper](#)